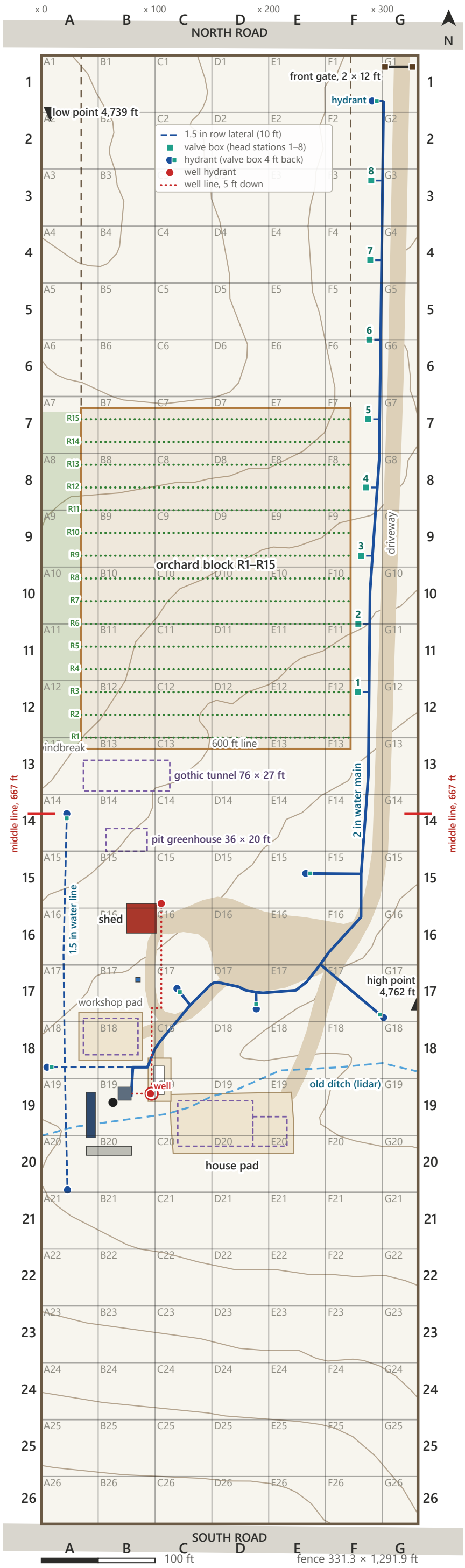
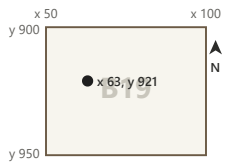
# Mountain Prairie Acres: Site Master Grid

One coordinate system for every project on the property: lot feet from the northwest fence corner, 50 ft cells A–G by 1–26, tied to latitude and longitude. Pages and code helpers all read the same data file.

### Whole lot, north up, 50 ft cells




## Reading a grid reference




**Lot feet:** *x* is feet east of the west fence, *y* is feet south of the north fence. The origin (0, 0) is the northwest fence corner, and the fences run true north–south and east–west.
**Cells:** a column letter (A–G, west to east) plus a row number (1–26, north to south). Each cell is 50 × 50 ft, so *B19* is x 50–100, y 900–950. Write a spot as *B19 (x 63, y 921)*.
**SiteRef:** `B19-E13-S21` means 13 ft east and 21 ft south of B19's northwest corner. See the SiteRef guide and examples on the second printed page.

## Lot and location

FENCED LOT
**331.3 × 1,291.9 ft**
9.83 acres inside the fence

LIKELY PARCEL
**330 × 1,320 ft (10 ac)**
to the road centerlines; not surveyed

ORIGIN (0, 0)
**43.47545, -115.59484**
northwest fence corner (NAD83 / GPS)

DRIVEWAY ENTRANCE
**43.47550, -115.59364**
open in Google Maps (for directions)

LOT AXES
**x runs 89.9° true**
within 0.2° of due east; y is due south

POSITION ACCURACY
**about 10 ft**
USGS NAIP-based; a shed GPS pin checks within 5 ft

## Feature coordinates and SiteRefs

SiteRefs below identify feature centers, not the full extent of a building or water line. Generated codes round to 0.01 ft; this does not improve the map's roughly 10 ft GPS accuracy.

| Feature (id) | Cells | SiteRef (center) | Size, ft | Ground, ft | Lat, long |
|---|---|---|---|---|---|
| **Boundary and access** | | | | | |
| Fence (lot line) `fence` | A1–G26 | `D13-E15.65-S45.95` | 331.3 × 1,291.9 | — | 43.47368, -115.59421 |
| Middle line (owner's tape) `lot_midline` | A14 → G14 | `D14-E15.65-S17` | 331.3 long | 4,755.5 | 43.47362, -115.59421 |
| North road `road_north` | off grid | `OUTSIDE LOT` | 24 wide | — | 43.47550, -115.59420 |
| South road `road_south` | off grid | `OUTSIDE LOT` | 24 wide | — | 43.47187, -115.59419 |
| Driveway entrance `driveway_entrance` | off grid | `OUTSIDE LOT` | — | — | 43.47550, -115.59364 |
| Front entrance gate `front_gate` | G1 | `G1-E14.3-S10` | 24 (2 × 12) | 4,748.7 | 43.47543, -115.59365 |
| 9 · Driveway `driveway` | G1 → E19 | `G10-E6.03-S5.3` | 14 wide | 4,752.8 | 43.47420, -115.59368 |
| 9 · Yard and turnaround `yard` | C15–F17 | `D16-E42.58-S30.35` | 8,400 sq ft | 4,759.0 | 43.47331, -115.59411 |
| 9 · Loop drive around the island `loop_path` | C16 → D17 | `C17-E1.54-S39.05` | 9 wide | 4,758.4 | 43.47315, -115.59445 |
| 9 · Path to the workshop pad `path_workshop` | C17 → B18 | `C18-E1.13-S5.25` | 10 wide | 4,758.8 | 43.47311, -115.59445 |
| 9 · Path to the RV pad `path_rv` | C18 | `C18-E3-S15.57` | 9 wide | 4,759.2 | 43.47308, -115.59444 |
| **Homestead** | | | | | |
| 1 · Red shed `shed` | B15–C16 | `B16-E38.25-S9` | 26.5 × 26 | 4,756.2 | 43.47337, -115.59450 |
| 2 · Outhouse `outhouse` | B17 | `B17-E35-S13` | 4 × 4 | 4,757.7 | 43.47322, -115.59451 |
| 3 · Solar container `container_solar` | A19, A20 | `A19-E43.5-S32` | 8 × 40 | 4,757.9 | 43.47290, -115.59467 |
| 3 · Storage container `container_storage` | A20, B20 | `B20-E9.5-S13.5` | 40 × 8 | 4,758.3 | 43.47281, -115.59461 |
| 4 · House pad `house_pad` | C19–E20 | `D19-E18.2-S39.45` | 109 × 54 | 4,759.8 | 43.47288, -115.59420 |
| 5 · Irrigation tank `irrigation_tank` | B19 | `B19-E13-S21` | 8 across | 4,758.7 | 43.47293, -115.59459 |
| 6 · Pump house `pump_house` | B19 | `B19-E23.25-S13.25` | 11.5 × 11.5 | 4,759.1 | 43.47295, -115.59456 |
| 7 · Workshop pad `workshop_pad` | A17–B18 | `B18-E11-S13` | 56 × 42 | 4,758.4 | 43.47308, -115.59460 |
| 8 · RV pad `rv_pad` | B18–C19 | `C19-E4-S1` | 20 × 38 | 4,759.5 | 43.47298, -115.59444 |
| 8 · RV camper `rv` | B18–C19 | `C19-E3.5-S1.5` | 9 × 25 | 4,759.5 | 43.47298, -115.59444 |
| **Planned buildings** | | | | | |
| 4 · House (planned) `house` | C19–D20 | `D19-E3-S39.5` | 66 × 40 | 4,759.6 | 43.47288, -115.59425 |
| 4 · Garage (planned) `garage` | D19–E20 | `E19-E1-S46.5` | 30 × 26 | 4,760.2 | 43.47286, -115.59407 |
| 7 · Workshop (planned) `workshop` | A17–B18 | `B18-E11-S13` | 48 × 32 | 4,758.4 | 43.47308, -115.59460 |
| Gothic tunnel greenhouse (planned) `greenhouse_gothic` | A13–C13 | `B13-E25-S33.5` | 76 × 27 | 4,752.4 | 43.47371, -115.59455 |
| Pit greenhouse (planned) `greenhouse_pit` | B14 | `B14-E25-S40` | 36 × 20 | 4,754.0 | 43.47356, -115.59455 |
| **Utilities** | | | | | |
| Water main, 2 in PVC `water_main` | B19 → G1 | `F11-E38.41-S46.24` | 2 in pipe, 1,015 long | 4,755.4 | 43.47396, -115.59375 |
| Water line west to the fence, 1.5 in PVC `water_tee_west` | A18, B18 | `A18-E42.75-S40` | 1.5 in pipe, 75 long | 4,758.5 | 43.47301, -115.59467 |
| Water line along the west fence, 1.5 in `water_line_west` | A14 → A20 | `A17-E20-S32.5` | 1.5 in pipe, 330 long | 4,756.9 | 43.47317, -115.59476 |
| Water branch to the island hydrant, 2 in `water_branch_island` | C17 | `C17-E25.1-S28.1` | 2 in pipe, 20 long | 4,758.7 | 43.47318, -115.59436 |
| Water branch to the hydrant south of the yard, 2 in `water_branch_yard_south` | D17 | `D17-E38.95-S31.35` | 2 in pipe, 15 long | 4,759.9 | 43.47317, -115.59412 |
| Water branch to the east hydrant, 2 in `water_branch_east` | E16 → G17 | `F17-E23.2-S22.95` | 2 in pipe, 75 long | 4,760.9 | 43.47320, -115.59380 |
| Water branch to the hydrant north of the yard, 2 in `water_branch_yard_north` | E15, F15 | `F15-E6.95-S19.8` | 2 in pipe, 50 long | 4,758.9 | 43.47348, -115.59386 |
| Water lateral at the north end, 1.5 in `water_lateral_north` | F1, G1 | `F1-E46-S40` | 1.5 in pipe, 10 long | 4,748.5 | 43.47534, -115.59372 |
| Hydrant, west end of the 1.5 in line `hydrant_west` | A18 | `A18-E5-S40` | — | 4,757.8 | 43.47301, -115.59481 |
| Hydrant, south end of the west fence line `hydrant_fence_south` | A20 | `A20-E23-S48` | — | 4,757.6 | 43.47271, -115.59474 |
| Hydrant, north end of the west fence line `hydrant_fence_north` | A14 | `A14-E22.4-S17` | — | 4,751.9 | 43.47362, -115.59475 |
| Hydrant, in the island `hydrant_island` | C17 | `C17-E19.3-S20.8` | — | 4,758.6 | 43.47320, -115.59438 |
| Hydrant, south of the yard `hydrant_yard_south` | D17 | `D17-E39.1-S38.7` | — | 4,760.0 | 43.47315, -115.59412 |
| Hydrant, near the east fence `hydrant_east` | G17 | `G17-E1-S46.3` | — | 4,761.5 | 43.47313, -115.59370 |
| Hydrant, north of the yard `hydrant_yard_north` | E15 | `E15-E32.6-S19.6` | — | 4,758.5 | 43.47348, -115.59396 |
| Hydrant, north end `hydrant_north` | F1 | `F1-E41-S40` | — | 4,748.3 | 43.47534, -115.59374 |
| *8 row laterals (1.5 in, 10 ft) with their head-station valve boxes, and 7 hydrant valve boxes: see Row laterals and valve boxes, below.* | | | | | |
| **Orchard** | | | | | |
| Orchard strip `orchard_area` | A1–F12 | `D7-E3.5-S0` | 237 × 600 | 4,744.1 | 43.47463, -115.59426 |
| Orchard block R1-R15 `orchard_block` | A7–F13 | `D10-E3.5-S10` | 237 × 300 | 4,750.8 | 43.47419, -115.59426 |
| Windbreak `windbreak` | A7–A13 | `A10-E17.5-S10` | 35 × 292 | 4,748.3 | 43.47419, -115.59477 |
| **Terrain** | | | | | |
| High point `high_point` | G17 | `G17-E31-S35` | — | 4,761.7 | 43.47316, -115.59358 |
| Low point `low_point` | A2 | `A2-E0-S0` | — | 4,739.2 | 43.47532, -115.59484 |
| Old ditch line `old_ditch` | A20 → G18 | `D19-E14.79-S11.83` | 8 wide | 4,759.6 | 43.47295, -115.59421 |

## Row laterals and valve boxes

Eight 10 ft, 1.5 in laterals (`water_lateral_row_1` to 8) run west off the 2 in main, each ending in a 1 in riser into a valve box (a row head station), numbered south to north on the map. Their spacing along the main is read from the owner's drone photo (October 2026): the five in the south are about 60 ft apart and the three in the north about 70 ft apart. The hydrants' valve boxes (4 ft back on the hydrant's line, where it has one) follow the head stations.

| Feature (id) | Cells | SiteRef (center) | Size, ft | Ground, ft | Lat, long |
|---|---|---|---|---|---|
| **Row head stations: the valve box at the end of each row lateral, south to north** | | | | | |
| Head station 1, valve box `head_station_1` | F12 | `F12-E28.3-S10` | 1 in riser, 10 ft lateral | 4,755.6 | 43.47392, -115.59379 |
| Head station 2, valve box `head_station_2` | F11 | `F11-E28.8-S0` | 1 in riser, 10 ft lateral | 4,753.7 | 43.47408, -115.59378 |
| Head station 3, valve box `head_station_3` | F9 | `F9-E31.3-S40` | 1 in riser, 10 ft lateral | 4,751.7 | 43.47425, -115.59378 |
| Head station 4, valve box `head_station_4` | F8 | `F8-E35.5-S30` | 1 in riser, 10 ft lateral | 4,749.7 | 43.47441, -115.59376 |
| Head station 5, valve box `head_station_5` | F7 | `F7-E37.6-S20` | 1 in riser, 10 ft lateral | 4,748.6 | 43.47458, -115.59375 |
| Head station 6, valve box `head_station_6` | F6 | `F6-E38.5-S0` | 1 in riser, 10 ft lateral | 4,750.3 | 43.47477, -115.59375 |
| Head station 7, valve box `head_station_7` | F4 | `F4-E39.3-S30` | 1 in riser, 10 ft lateral | 4,749.6 | 43.47496, -115.59375 |
| Head station 8, valve box `head_station_8` | F3 | `F3-E40.2-S10` | 1 in riser, 10 ft lateral | 4,748.9 | 43.47515, -115.59374 |
| **Hydrant valve boxes, 4 ft back on the hydrant's line** | | | | | |
| Valve box, west end of the 1.5 in line `valve_box_hydrant_west` | A18 | `A18-E9-S40` | 4 ft back on the line | 4,757.9 | 43.47301, -115.59480 |
| Valve box, north end of the west fence line `valve_box_hydrant_fence_north` | A14 | `A14-E22.3-S21` | 4 ft back on the line | 4,752.0 | 43.47361, -115.59475 |
| Valve box, in the island `valve_box_hydrant_island` | C17 | `C17-E21.8-S23.9` | 4 ft back on the line | 4,758.6 | 43.47319, -115.59437 |
| Valve box, south of the yard `valve_box_hydrant_yard_south` | D17 | `D17-E39-S34.7` | 4 ft back on the line | 4,759.9 | 43.47316, -115.59412 |
| Valve box, near the east fence `valve_box_hydrant_east` | F17 | `F17-E47.9-S43.7` | 4 ft back on the line | 4,761.5 | 43.47314, -115.59371 |
| Valve box, north of the yard `valve_box_hydrant_yard_north` | E15 | `E15-E36.6-S19.6` | 4 ft back on the line | 4,758.6 | 43.47348, -115.59394 |
| Valve box, north end `valve_box_hydrant_north` | F1 | `F1-E45-S40` | 4 ft back on the line | 4,748.5 | 43.47534, -115.59373 |

## SiteRef Location Codes and Shared Helpers

**Format:** `CELL-Eoffset-Soffset` fits a precise location in one printed column. `C17-E12-S18` means 12 ft east and 18 ft south of **C17's northwest corner**: x 112, y 818. Decimals are feet, not inches; `C17-E12.5-S18` is valid.
**Planted example:** the M-111 rootstock at orchard R6-C3 is `B11-E33-S0` (x 83, y 500) in the current layout. SiteRef locates the spot, not the plant: keep the metal tag and individual Plant ID separately. A replacement plant can reuse the spot's code.
**Unknown locations:** record `PENDING`, never guessed offsets. The three Urban apples' outside-orchard locations are pending. Keep cell and offsets as underlying values and generate SiteRef for printing; do not maintain a second editable copy.
**GPS input:** the shared metadata stores one latitude/longitude tie for the lot. Each cell's northwest-corner GPS coordinates are calculated from its x0, y0 values, not stored as a second table. Convert a GPS point with `SiteGrid.fromLatLon(latitude, longitude)`, then generate its SiteRef. Latitude comes first; Idaho longitudes are negative. Near cell boundaries, phone GPS error plus the roughly 10 ft map accuracy can put a point in the neighboring cell; verify measured offsets.
Web pages load generated `data/site_grid.js` and `data/site_terrain.js`, then call `SiteGrid`. Edit canonical JSON in the repository's `data/` directory; see the property data and SQLite guide.

```
<script src="data/site_grid.js"></script>
<script src="data/site_terrain.js"></script>

SiteGrid.cellOf(63, 921)                    // 'B19'
SiteGrid.cellBounds('B19')                  // { x0: 50, y0: 900, x1: 100, y1: 950 }
SiteGrid.siteRef(112, 818)                  // 'C17-E12-S18'
SiteGrid.fromSiteRef('C17-E12-S18')         // [112, 818]
SiteGrid.fromSiteRef('PENDING')             // null (no location)
SiteGrid.siteRef(...SiteGrid.fromLatLon(43.473208398831, -115.594410245142))// 'C17-E12-S18' (GPS to
SiteRef)
SiteGrid.center('pump_house')               // [73.25, 913.25]
SiteGrid.cellsOf('house_pad')               // ['C19', 'D19', 'E19', 'C20', 'D20', 'E20']
SiteGrid.elevationFt(63, 921)               // 4758.7 (feet above sea level)
SiteGrid.toLatLon(63, 921)                  // [43.472925, -115.594594]
SiteGrid.fromLatLon(43.473383, -115.594507)  // [86.5, 754.3] (a GPS pin on the shed)
SiteGrid.treeXY(5, 3)                       // [74, 520] (orchard tree R5-C3, cell B11)
SiteGrid.mapsLink(63, 921)                  // Google Maps link to that spot
```

Scripts and the command line use `tools/site_grid.py`, which reads the same files:

```
python tools/site_grid.py cell 63 921             # B19
python tools/site_grid.py where B19               # cell bounds and the features in it
python tools/site_grid.py feature pump_house      # center, size, cells, ground, lat/long
python tools/site_grid.py dist pump_house house   # gap and ground change
python tools/site_grid.py tree 5 3                # orchard tree R5-C3: x, y, cell, ground
python tools/site_grid.py site-ref C17-E12-S18     # x 112, y 818, cell C17
python tools/site_grid.py site-ref PENDING        # location not yet known
python tools/site_grid.py fromlatlon 43.473383 -115.594507
from site_grid import SiteGrid; grid = SiteGrid()
grid.site_ref(83, 500)                            # B11-E33-S0 (R6-C3 M-111)
grid.from_site_ref("C17-E12-S18")                  # (112, 818)
grid.from_site_ref("PENDING")                     # None
grid.site_ref(*grid.from_latlon(43.473208398831, -115.594410245142))  # C17-E12-S18
```

## Conventions

- Feet throughout. Elevations are above sea level (NAVD88) from USGS 3DEP lidar flown before the homestead pads were graded.
- SiteRef uses the master cell, then east and south offsets from that cell's northwest corner. Offsets are nonnegative, less than 50 ft, and must stay inside the fence, including the partial edge cells. Generated codes use up to two decimal places. E and S are compass directions, even on the orchard map drawn south up.
- Planting records should identify each individual plant, not assign one coordinate to a quantity group. SiteRef is calculated from its location; it is not a metal tag or a permanent Plant ID. Keep unlocated plants PENDING and vacant spots separate from dead or moved plants.
- Status is existing or planned; planned buildings (the house, garage, workshop, and the two greenhouses) move once staked. The greenhouse outlines are the footprints from the greenhouse pages; the pit greenhouse's berm, retaining wall, and swales are on the greenhouse site plan, not here. Edit canonical data/site.json (rect x0/y0/x1/y1, circle cx/cy/r, polygon/line pts, point x/y), then run python tools/property_data.py build and the tests. Browser files in docs/data/ are generated, not independently editable.
- The red ticks on the east and west fences mark the owner's taped middle line (lot_midline): 667 ft south of the north fence, on the A14 hydrant and 79 ft north of the red shed's north building corner. Half the 1,291.9 ft fence is still 645.95 ft, 21 ft north of these ticks. A tape from the northwest fence corner would confirm the 667.
- The front gate (front_gate, cell G1): 2 × 12 ft metal leaves that swing both ways on treated junior railroad ties (about 8 × 12 in) set in concrete, drawn closed. Placed from the October 2026 drone photos and the owner: the east post is about 5 ft inside the east fence, the gate about 10 ft inside the mapped north fence, good to about 8 ft.
- Latitude and longitude come from the fitted lot frame, good to about 10 ft: fine for maps and directions, not for boundary questions. For precise layout, measure from the fences.
- Buried water lines (2 in PVC solid, 1.5 in dashed) and the frost-free hydrants (blue dots) are traced from the owner's hand-drawn map (October 2026): good to about 5 ft around the homestead and 10 ft at the north end, so dig by hand near them. The two well hydrants use that same dot in red, on the red dotted well line.
- The 8 row laterals (10 ft, 1.5 in, west off the 2 in main) end in a 1 in riser and a valve box (teal squares numbered 1–8, south to north). Their spacing is read from the owner's drone photo (October 2026), so the positions along the main are good to about 8 ft. 7 of the 10 hydrants also have a valve box 4 ft back on their line (none at A20, B19, C15): a small teal chip on the hydrant dot here, drawn apart on the homestead plot map, and listed under Row laterals and valve boxes.